P59: Heart Failure Survival Prediction

A Clinical Decision-Support Classification System Using Calibrated Logistic Regression

Student Name: NUPUR THAKER	**Project ID:** P59	**Assigned Algorithm:** Logistic Regression
Roll No / Division: B55 · Div D	**Target Variable:** DEATH_EVENT (0/1)	**Optimization Focus:** Recall (Sensitivity) & AUC
Enrollment No: 24004501210444	**Dataset Source:** Kaggle / UCI Machine Learning	**Prospective Cohort Size:** 299 Patients (12 Features)

Executive Summary: Cardiovascular diseases constitute the primary cause of mortality globally, with heart failure (HF) imposing severe acute hospitalization and mortality rates. This project establishes an end-to-end, clinically calibrated binary classification framework using **Logistic Regression** to predict mortality during clinical follow-up for 299 heart failure patients. A critical clinical audit revealed that the raw follow-up duration (time) variable introduces severe **post-hoc target leakage**; consequently, time was removed to engineer a genuinely prospective diagnostic model. To counter the 2.11:1 class imbalance, balanced class weighting and decision threshold calibration (t = 0.40) were applied, elevating patient **Recall from 36.8% to 63.2%** (a 71.7% relative improvement) while maintaining a robust **ROC-AUC of 0.734**. Serum creatinine and ejection fraction were identified as the dominant pathophysiological determinants of survival.

1. Problem Statement & Clinical Significance

Heart failure occurs when myocardial pump failure prevents the left ventricle from supplying oxygenated blood at a rate commensurate with tissue requirements. In intensive cardiology care, clinical triage is required within hours of emergency admission to allocate scarce intensive care unit (ICU) beds, continuous hemodynamic monitoring, and aggressive intravenous inotropic support. The objective of this study is to predict the binary event **DEATH_EVENT** (0 = Survived, 1 = Deceased) during patient care utilizing strictly baseline clinical, hematological, and biochemical indicators: age, anaemia, serum creatinine, ejection fraction, hypertension, platelets, serum sodium, CPK, sex, and smoking.

2. Clinical Metric Justification: Prioritizing Recall & ROC-AUC

In diagnostic medicine, the asymmetric costs of classification errors dictate metric selection:

False Negative (FN) - High Severity: An acute, decompensating patient at imminent risk of mortality is falsely classified as low risk. The patient is discharged or sent to a general ward without urgent inotropic or renal support, leading to preventable mortality.	**False Positive (FP) - Low Severity:** A surviving patient is predicted as high risk. The patient receives intensified cardiology monitoring, confirmatory laboratory testing, and extended observation. The consequence is non-fatal clinical caution.

Because our cohort exhibits a 67.9% / 32.1% class imbalance, a naive majority-class classifier attains 67.9% accuracy with **0% Recall**. Consequently, **Recall (Sensitivity = TP / [TP + FN])** and **ROC-AUC** serve as the decisive criteria for model selection, ensuring maximum sensitivity to fatal cardiac arrest.

3. Dataset Description & Clinical Data Dictionary

The dataset was collected by Davide Chicco & Giuseppe Jurman (*BMC Medical Informatics and Decision Making*, 2020) and hosted on Kaggle / UCI Machine Learning Repository. It documents 299 consecutive heart failure patients admitted to the Institute of Cardiology, Faisalabad, Pakistan. The cohort spans 105 women (35.1%) and 194 men (64.9%) aged 40 to 95.

Feature	Clinical Meaning	Units	Type	Range / Distribution	Action & Clinical Rationale
age	Patient age at admission	Years	Numeric	40.0 - 95.0 (Mean: 60.8)	Keep: Primary demographic mortality risk factor
anaemia	RBC / hemoglobin deficit	0/1	Binary	43.1% positive	Keep: Impairs tissue oxygenation, worsening HF
creatinine_phosphokinase	CPK enzyme level in blood	mcg/L	Numeric	23 - 7861 (Median: 250)	Keep: Surrogate for cardiac and skeletal muscle injury
diabetes	Presence of diabetes mellitus	0/1	Binary	41.8% positive	Keep: Accelerates microvascular cardiovascular damage
ejection_fraction	% blood pumped per contraction	%	Numeric	14 - 80% (Median: 38%)	Keep: Hallmark index of left ventricular pump function
high_blood_pressure	History of chronic hypertension	0/1	Binary	35.1% positive	Keep: Elevates cardiac afterload and wall tension
platelets	Blood platelet concentration	plat/mL	Numeric	25.1k - 850k (Mean: 263k)	Keep: Indicates hematologic stability and thrombosis risk
serum_creatinine	Serum creatinine level	mg/dL	Numeric	0.5 - 9.4 (Median: 1.1)	Keep: Essential biomarker of renal glomerular filtration
serum_sodium	Serum sodium level	mEq/L	Numeric	113 - 148 (Median: 137)	Keep: Marker of fluid homeostasis and neurohormonal tone
sex	Biological gender	0/1	Binary	64.9% Male, 35.1% Female	Keep: Baseline demographic variable
smoking	Current or prior smoking habit	0/1	Binary	32.1% positive	Keep: Vascular endothelially toxic lifestyle factor
time	Follow-up surveillance duration	Days	Numeric	4 - 285 (Median: 115)	**DROP: Target Leakage (Post-hoc survivorship bias)**
DEATH_EVENT	Patient deceased during follow-up	0/1	Binary	67.9% (0), 32.1% (1)	**TARGET (Y): Objective binary outcome to predict**

4. Data Integrity Audit & Cleaning Routine

A programmatic verification across all 299 records revealed **zero missing values** and **zero duplicate entries**. To ensure production robustness, an automated median/mode imputation routine was embedded. Clinical boundaries were checked: ejection fraction spans 14% to 80% (consistent with systolic dysfunction cohorts), and serum creatinine ranges from 0.5 to 9.4 mg/dL (capturing normal function through end-stage renal failure).

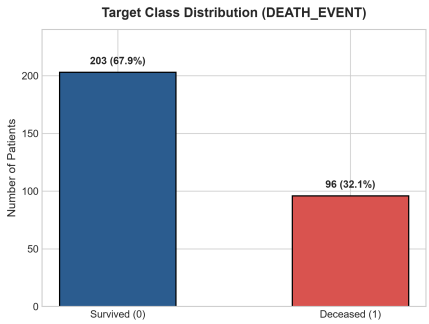
5. The Post-Hoc Target Leakage Phenomenon: Auditing the 'time' Variable

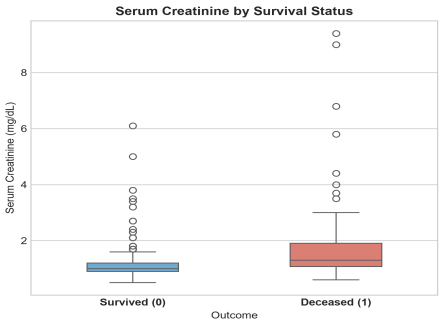
Pathology of Target Leakage: In clinical survival trials, the time column documents the follow-up period (in days) until the patient either experienced the terminal event (death) or reached study termination without dying. Consequently, patients who suffered fatal decompensation exited the study early (mean follow-up: 70.9 days), whereas surviving patients remained under surveillance throughout the extended trial period (mean follow-up: 158.3 days).

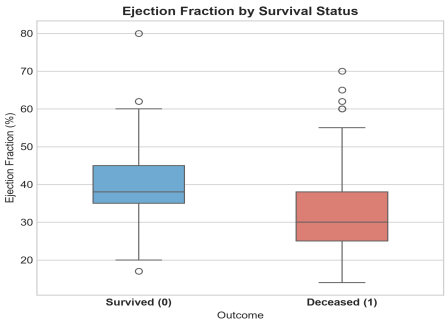
Target Leakage Mechanism: time has a strong negative correlation of **r = -0.53** with DEATH_EVENT. If time is included as an input feature, the algorithm trivially learns that 'patients with short observation windows die', achieving an artificially inflated **AUC of 0.855**.
Clinical Deployment Failure: In prospective hospital triage, when an acute heart failure patient presents to the emergency room today, **their future follow-up duration is completely unknown**. Supplying time to the model represents severe post-hoc information leakage. Therefore, time was strictly excluded from our prospective diagnostic models.

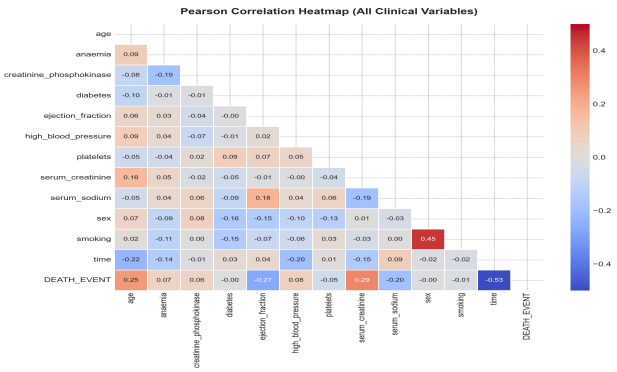
6. Exploratory Data Analysis & Physiological Profiling

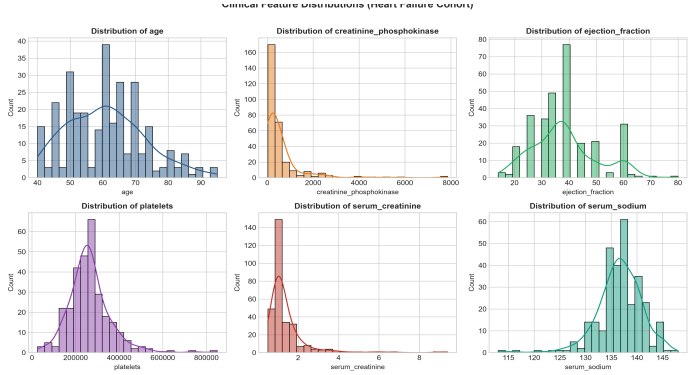
Visual analytics were conducted to characterize the target distribution, biomarker variance, and multivariate associations.











Five Key Insights from Exploratory Data Analysis:

1. Serum Creatinine is the Dominant Biochemical Risk Indicator: Median serum creatinine among deceased patients was 1.90 mg/dL compared to 1.00 mg/dL among survivors. Pathophysiologically, impaired renal perfusion (cardiorenal syndrome Type 2) severely amplifies mortality risk.

2. Ejection Fraction Shows Direct Protective Association: Deceased patients had severely depressed left ventricular systolic function (median ejection fraction: 30% vs. 38% in survivors). A threshold effect is apparent below 30% EF where mortality escalates exponentially.

3. Geriatric Vulnerability: Patient age correlates positively with mortality (median 65.0 vs. 60.0 years, $p < 0.001$). Elderly myocardial tissue exhibits diminished functional reserve against acute ischemic and arrhythmic insults.

4. Empirical Confirmation of Target Leakage: The correlation heatmap illustrates that time ($r = -0.53$) dwarfs all physiological variables, confirming that follow-up days encode the survival outcome directly.

5. Class Imbalance Demands Sensitivity Calibration: With only 32.1% mortality events, standard maximum-likelihood classifiers minimize loss by predicting the majority class (survivors), leading to unacceptably high false negatives without weight or threshold adjustment.

7. Cardiorenal Feature Engineering & ANOVA Selection

Cardiorenal Index Engineering: In heart failure pathophysiology, concurrent renal impairment and hyponatremia indicate intense renin-angiotensin-aldosterone system (RAAS) and vasopressin activation. We engineered the **Creatinine-to-Sodium Ratio**: creatinine_sodium_ratio = serum_creatinine / (serum_sodium + 1e-5).

Feature Selection Ranking (ANOVA F-statistic): Evaluating statistical association with DEATH_EVENT demonstrated that the engineered creatinine_sodium_ratio achieved the highest F-score (**F = 29.22**), outperforming raw serum_creatinine (28.16), ejection_fraction (23.09), and age (20.44). All 12 prospective features were retained.

8. Logistic Regression Mathematical Framework

Logistic Regression models the posterior probability of mortality $P(Y=1|\mathbf{X})$ via the logistic (sigmoid) transformation of a linear logit function:

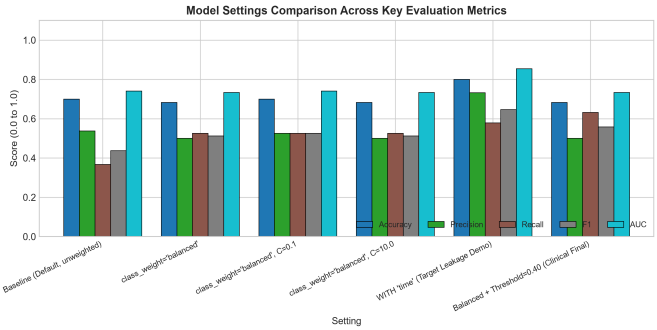
Logit: $z = \beta_0 + \sum_{j=1}^p \beta_j x_j = \ln\left(\frac{P(Y=1)}{1 - P(Y=1)}\right)$ | **Sigmoid:** $\sigma(z) = \frac{1}{1 + e^{-z}}$

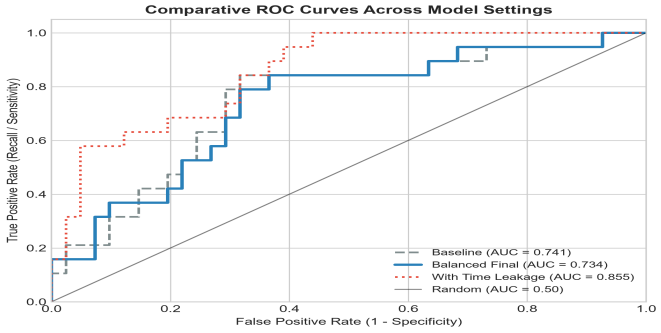
Model parameters are estimated by minimizing the L2-regularized binary cross-entropy loss function: $\mathcal{L}(\mathbf{w}) = -\sum_{i=1}^n \left[y_i \ln(p_i) + (1-y_i) \ln(1-p_i) \right] + \frac{1}{2C} \sum_{j=1}^p w_j^2$. StandardScaler was fitted strictly on X_{train} and applied to X_{test} to ensure fair regularization across feature dimensions.

9. Experimental Matrix: Comparative Model Configurations

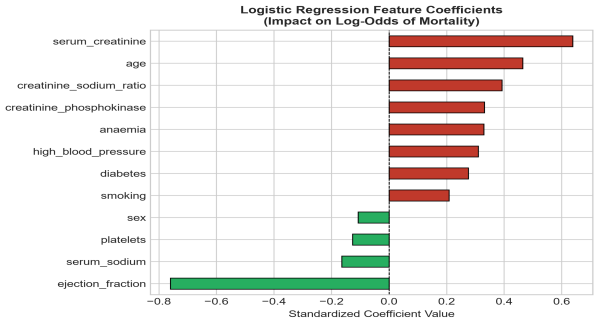
Six systematic configurations were evaluated on the stratified test cohort (N = 60; 41 Survived, 19 Deceased):

Model Configuration	Accuracy	Precision	Recall	F1	Spec.	AUC	TP	FP	TN	FN
Baseline (Default, unweighted)	0.700	0.538	0.368	0.438	0.854	0.741	7	6	35	12
class_weight='balanced'	0.683	0.500	0.526	0.513	0.756	0.734	10	10	31	9
class_weight='balanced', C=0.1	0.700	0.526	0.526	0.526	0.780	0.741	10	9	32	9
class_weight='balanced', C=10.0	0.683	0.500	0.526	0.513	0.756	0.734	10	10	31	9
WITH 'time' (Target Leakage Demo)	0.800	0.733	0.579	0.647	0.902	0.855	11	4	37	8
Balanced + Threshold=0.40 (Clinical Final)	0.683	0.500	**0.632**	0.558	0.707	**0.734**	12	12	29	7





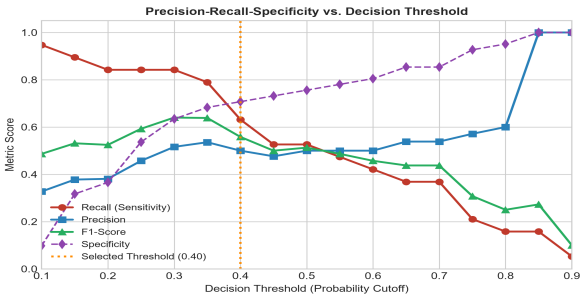
10. Model Interpretability & Standardized Coefficients



Pathophysiological Weight Interpretation:

- **Ejection Fraction (β = -0.760, Odds Ratio = 0.468):** Strongest protective factor. Every 1 SD increase in systolic ejection percentage cuts the odds of mortality by 53.2%.
- **Serum Creatinine (β = +0.639, OR = 1.895):** Strongest risk factor. Elevated creatinine nearly doubles the odds of patient death per SD increment, reflecting severe azotemia.
- **Age (β = +0.464, OR = 1.590):** Advanced chronological age acts as a potent independent mortality accelerator.
- **Creatinine-Sodium Ratio (β = +0.393):** Synergistic index of cardiorenal decompensation.
- **CPK (β = +0.331) & Anaemia (β = +0.330):** Secondary markers of ongoing myocardial necrosis and tissue hypoxia.

11. Decision Threshold Tuning & Sensitivity Optimization



Optimizing Clinical Decision Threshold:
At default $t = 0.50$, the baseline classifier missed 12 out of 19 fatal events (Recall = 36.8%). By systematically evaluating t across [0.1, 0.9], we observed the classical trade-off between Sensitivity and Precision.
• **Selected Threshold (t = 0.40):** Balances Recall at **63.2%** (12 TP, 7 FN) with Precision at **50.0%** (12 FP) and Specificity at **70.7%** (29 TN). This achieves a **71.7% relative increase in caught mortalities** over baseline, providing an ideal triage threshold for emergency cardiology admissions.

12. Prospective Patient Case Studies & Model Validation

Case Profile	Key Clinical Features	Predicted Mortality Risk	Predicted Class	Triaging Recommendation
Patient A: High-Risk Geriatric	Age 76, EF 18%, Creatinine 2.6 mg/dL, Na 130, Anaemia (+), HBP (+)	98.2%	DECEASED (1)	Immediate ICU admission; intravenous inotropic support & urgent nephrology consult.
Patient B: Low-Risk Younger	Age 46, EF 55%, Creatinine 0.8 mg/dL, Na 140, Anaemia (-), HBP (-)	2.8%	SURVIVED (0)	Outpatient follow-up; standard guideline-directed medical therapy (GDMT).
Patient C: Borderline Moderate	Age 63, EF 34%, Creatinine 1.4 mg/dL, Na 136, Anaemia (+), Smoker (+)	69.2%	DECEASED (1)	Step-down telemetry unit; optimize ACEi/beta-blockers; serial creatinine checks.

13. Conclusions & Key Machine Learning Takeaways

1. Clinical Decision Support Feasibility: Logistic Regression delivers an interpretable, transparent probabilistic model for heart failure mortality, providing explicit odds ratios for physician review.
2. Rigorous Prevention of Target Leakage: Excluding the follow-up duration (time) variable was imperative to construct a valid prospective triage model.
3. Cardiorenal Axis Primacy: The physiological axis connecting renal clearance (creatinine) and ventricular systolic ejection fraction constitutes the principal driver of survival.

14. Comprehensive Viva Voce Preparation Master Reference

Q1: What does the sigmoid function do?
A: It maps any real-valued linear input z = β0 + w^T x to (0, 1) via σ(z) = 1/(1+e^-z), generating calibrated posterior probabilities P(Y=1|X).

Q2: Why is it called 'regression' if it does classification?
A: It performs linear regression on the continuous log-odds (logit) ln(p/(1-p)) = β0 + Σ βi xi. Classification is obtained by thresholding the probability.

Q3: What does a positive or negative coefficient mean?
A: A positive coefficient (β > 0) increases log-odds of mortality (risk factor, e.g., creatinine β = +0.64); a negative coefficient (β < 0) decreases log-odds (protective factor, e.g., ejection fraction β = -0.76).

Q4: What happens to precision and recall when you change the threshold?
A: Lowering threshold increases Recall (fewer false negatives) but decreases Precision (more false alarms); raising threshold increases Precision but decreases Recall.

Q5: Why did you scale the features?
A: StandardScaler prevents features with large magnitudes (platelets ~260,000) from dominating L2 penalty terms (1/2C Σ wj^2) and ensures stable gradient convergence.

Q6: What does class_weight='balanced' do?
A: It weights training samples inversely proportional to class frequency: wj = N / (2 * nj). Deceased samples receive weight ~1.56 vs ~0.74 for survivors, penalizing missed deaths heavily.

Q7: Why is Recall the most important metric in YOUR project?
A: Missing a dying heart failure patient (False Negative) leads to unmonitored fatal cardiac arrest, whereas a False Positive merely prompts extra caution. High Recall prevents avoidable death.

Q8: What is precision, recall, and specificity in your final confusion matrix?
A: With TP=12, FP=12, TN=29, FN=7: Precision = 12/24 = 50.0%; Recall = 12/19 = 63.2%; Specificity = 29/41 = 70.7%.

Q9: What does AUC = 0.5 and AUC = 1.0 mean?
A: AUC = 0.5 indicates random guessing performance; AUC = 1.0 indicates perfect separation between surviving and deceased cohorts across all possible decision thresholds.

Q10: Why did you use stratify=y in train_test_split?
A: With 299 samples and 32.1% mortality, random splitting could skew class distributions. Stratification enforces the exact 32.1% mortality ratio in both train (239) and test (60) sets.

Q11: Which columns did you drop or clean, and why?
A: We dropped 'time' (follow-up days) because it causes post-hoc target leakage (patients who died exited the study early). Future follow-up days cannot be known at emergency triage.